THE UNIVERSITY OF MANCHESTER

PARTICULARS OF APPOINTMENT

FACULTY OF SCIENCE & ENGINEERING

SCHOOL OF ENGINEERING

DEPARTMENT OF CHEMICAL ENGINEERING

RESEARCH ASSOCIATE IN SOLID-STATE NMR SPECTROSCOPY

VACANCY REF: SAE-032399

Salary:	UoM Grade 6 £37,694 - £46,049 per annum depending on relevant experience
Hours:	Full time (1 FTE)
Duration:	Fixed Term for 36 months
Location:	Oxford Road, Manchester

Enquiries about the vacancy, shortlisting, and interviews:
Name: Dr Daniel Lee
Contact: daniel.lee@manchester.ac.uk

Background

This Postdoctoral Research Associate (PDRA) position is part of a new £7m EPSRC funded Programme Grant that brings together a team of researchers from the universities of Durham, Leeds and Manchester investigating the formation of Molecular Solid Solutions (MoSS).

Solid solutions are created when atoms or ions of a dopant are incorporated within the crystal structure of a host. Even at minute quantities, this can significantly alter the properties of crystals. Although solid solutions in metals and inorganic materials are well-studied, little is known about systems where the host and dopants are small organic molecules (MoSS). There is huge potential for using MoSS to change and modulate a material's properties, such as solubility, crystal polymorph, crystal sizes and shapes, and rates of dissolution.

The overall goal of the consortium is to develop and apply cutting-edge techniques in crystal synthesis, characterisation, and modelling to study and generate a new understanding of MoSS. These advances will have applications across multiple sectors, including pharmaceuticals, agrochemicals, and food industries. There will be multiple opportunities to collaborate, take part in network events, and undertake research placements within this project.

The PDRA will have a PhD in the physical sciences or a closely related field and will have a background in solid-state nuclear magnetic resonance (ssNMR) spectroscopy. The key responsibilities are to develop, test, and implement different experimental methodologies to gain structural insights to help characterise and predict MoSS formation.

Host institution

The University of Manchester (www.manchester.ac.uk) is one of the largest single-site universities in the UK, with one of the biggest student communities. In total, 25 Nobel Prize winners have worked or studied here and 83% of our research was ranked as 'world-leading' or 'internationally excellent' by the Research Excellence Framework in 2014. Furthermore, more than nine out of ten of our recent graduates go straight into employment or continued studies. The Faculty of Science and Engineering (FSE) comprises two multi-discipline Schools; the School of Engineering and the School of Natural Sciences, each led by a Head of School and Head of School Operations. The School of Engineering is made up of four academic departments and the School of Natural Sciences is made up of five academic departments. For more information please visit https://www.se.manchester.ac.uk/.

FSE has the full range of analytical facilities required to support globally competitive research. The NMR service has 16 spectrometers, ranging from a 300 MHz machine dedicated to undergraduate teaching through to a recently purchased liquid helium-cooled cryoprobe, high-sensitivity 700 MHz instrument and biomolecular dedicated 800 MHz instrument. There are also NMR instruments dedicated to radioactive samples and to solids. The UK DNP MAS NMR Facility at Manchester complements these services and will be used as part of this project.

Overall Purpose of the Job

The PDRA role will involve the synthesis and investigation of MoSS. As well as ssNMR spectroscopy, complementary characterisation techniques and computational modelling will be employed where deemed necessary. The PDRA will work closely with other researchers from the MoSS Programme Grant.

Key Responsibilities, Accountabilities, or Duties

The range of duties will include:

- Conduct individual and collaborative research projects according to the overall needs of the
- research program.
- Develop research objectives and proposals for own or joint research.
- Write up research work for publication.
- Continually update knowledge and understanding in field or specialism.
- Translate knowledge of advances in the subject area into research activity.
- Deal with routine communication using a range of media.
- Communicate complex information, orally, in writing, and electronically.
- Liaise with colleagues and students.
- Work with colleagues on joint projects, as required.
- Collaborate with academic colleagues on areas of shared research interest.
- Attend and contribute to relevant project meetings.
- Use new research techniques and methods.
- Use creativity to analyse and interpret research data and draw conclusions on the outcomes.
- Contribute to collaborative decision making with colleagues in areas of research.
- Use research resources, laboratories, and workshops as appropriate.
- Plan and manage own research activity in collaboration with others.
- Balance the competing pressures of research and administrative demands and deadlines.
- Be aware of the risks in the work environment and their potential impact on their own work and

that of others.
- Post holders are required to familiarise themselves with the University's Equality and Diversity policies and to actively support these wherever possible.

PERSON SPECIFICATION

Essential Knowledge, Skills, and Experience:

- A PhD in the physical sciences or related subject.
- A degree in chemistry, materials, chemical engineering, or similar, with practical laboratory experience or extensive work experience in a similar role.
- Experience in implementing a wide range of advanced solid-state NMR experiments.
- Experience with multinuclear solid-state NMR spectroscopy (e.g. ^{1}H, ^{2}H, ^{13}C, ^{15}N, ^{17}O, ^{19}F and other nuclei as appropriate).
- Good data analysis skills, including processing and simulation of NMR data.
- Excellent communication and interpersonal skills.
- Ability to work both independently and collaboratively as a member of a multi-disciplinary team.
- Excellent time management and organisational skills.
- Ability to liaise confidently and effectively with a range of individuals.
- Flexible approach to dealing with research problems as they arise.
- Willingness to learn and develop.
- Ability to present in both written and oral publications.
- Ability to meet deadlines.
- Strong record of journal publications appropriate to career stage.
- An awareness and understanding of relevant health and safety requirements, including the application of risk assessments and operating procedures etc. in a laboratory environment.
- Ability to assess and organise resources.
- Contribution to the supervision and support of undergraduate, Master's, or PhD students.
- A flexible approach to working with the willingness to travel and work outside normal business hours (if occasionally required).

Desirable Knowledge, Skills, and Experience:

- Practical experience running a DNP MAS NMR spectrometer and associated experiments.
- Practical experience with computational modelling (e.g. spin dynamics simulations, molecular dynamics, DFT).
- Experience correlating NMR data with structural information derived from diffraction or computational methods.
- Experience in the characterisation of molecular solids and/or crystalline organic materials.
- Experience with complementary characterisation methods such as PXRD, DSC, TGA, microscopy, Raman, IR spectroscopy, or solid-form analysis.
- Experience of working with industrial and academic partners.
- Experience presenting research findings at national and/or international conferences.